

· Callie Clontz ·

POSTDOCTORAL RESEARCHER & OMEGACAT COLLABORATION MEMBER

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Education

Postdoctoral Researcher

Heidelberg, Germany

MAX PLANCK INSTITUTE FOR ASTRONOMY

Since Aug. 2025

- Advisor: Prov. Doc. Dr. Nadine Neumayer (neumayer@mpia.de, +49 6221 528 446)
- Research: Combined photometry + metallicity to parse individual stellar populations in Omega Centauri, linking probes of nuclear star cluster assembly. Obtained follow-up high resolution spectroscopic abundances, analysis ongoing.

PhD in Physics, Concentration in Astronomy

UT, USA + Heidelberg, Germany

UNIVERSITY OF UTAH

Aug. 2021 - Aug. 2025

- Advisor: Prof. Dr. Anil Seth (aseth@astro.utah.edu, +1 801-581-7200)
- Research: Uncovered the insitu vs. accreted contributions to the assembly history of Omega Centauri via the age-metallicity relation, and helium enhancement as a function of [Fe/H].

MS in Physics, Concentration: Astronomy

Salt Lake City, UT USA

UNIVERSITY OF UTAH

Aug. 2019 - Aug. 2021

- Advisor: Prof. Dr. Zheng Zheng (zhengzheng@astro.utah.edu)
- Research: Modeled statistical galaxy clustering analyses to quantify central and satellite host halo occupation rates, at intermediate redshift, via Dark Matter Halo Occupation Distribution Modeling.

BS in Physics, Minor in Mathematics

Asheville, NC USA

UNIVERSITY OF NORTH CAROLINA AT ASHEVILLE

Jan. 2017 - May 2019

- Advisor: Dr. David Wake
- Research: Measured galaxy clustering environment dependence on H-alpha luminosity and stellar mass via 2-point correlation function.

Relevant Skills

Programming languages, Python, C++, R, MATLAB, Unix/Linux

Tools, LaTeX, Jupyter, DS9, VSCode, vim, Git, cluster computing, TOPCAT

Libraries, ESOREflex, PPXF, TSPy, sci-kit learn, multiprocessing, astropy, scipy, matplotlib, joblib

Methods, spectroscopic data reduction, spectroscopic abundance measurements, MCMC, Maximum Likelihood, Bayesian Inference, Machine Learning: [Preprocessing, Classification, Regression, Clustering, Dimensionality-Reduction (PCA), Supervised (Nearest Neighbor), Unsupervised (GMM)], direct observations, empirically derived photometric corrections, deconvolution, 2-pt Correlation, Cross-correlation, Halo Occupation Distribution modeling, 2D-Gaussian fitting, machine readable tables, data cleaning

Teaching and Mentoring Experience

2024 - 2026	Co-advising a Master's Student , My student analyzed unique features in the MUSE spectra of the anomalous metal rich population of Omega Centauri.	MPIA
2020-2021	Mentor Teaching Assistant , Organized and hosted new TA orientation (week-long intensive training), provided teaching observations + feedback (2 semesters)	U of U
2019-2021	Teaching Assistant , Discussion Section, Physics I and II (4 semesters): Provided conceptual review, facilitated group discussion, created + presented solutions, tutored, graded	U of U
2018-2019	Docent , Lookout Observatory: Hosted public and private events, guided deep imaging and by-eye planet viewings	UNCA
2018-2019	Peer Study Group Facilitator , (4 semesters): Mediated small group "productive struggle"	UNCA
2017-2019	Physics Tutor , (4 semesters): Helped students utilize resources and enhance their problem solving strategies	UNCA

Additional Experience & Awards

Successful ESO P115 Proposal

Paranal Observatory, Chile

UT2: FLAMES/GIRAFFE

April 2025

- Obtained 13 hours telescope time, observing 130 targets across both streams of ω Cen's age-metallicity relation.

Telescope Time Allocation: Distributed Peer Reviewer

EUROPEAN SOUTHERN OBSERVATORY

Heidelberg, Germany

Oct. 2024

- Reviewed, commented, and ranked 10 proposals related to star clusters and stellar populations.

Manuscript Referee

ASTRONOMY AND ASTROPHYSICS JOURNAL

Heidelberg, Germany

July 2025

- Provided comment on submitted paper to improve scientific results and the clarity of their results.

Visiting Researcher Stay

MAX PLANCK INSTITUTE FOR ASTRONOMY

Heidelberg, Germany

2022-2025

- Collaborated in-person with the oMEGACat team, resulting in 7 co-authored publications so far.

Swigart Graduate Fellowship

UNIVERSITY OF UTAH

Salt Lake City, UT USA

2020

- Proposed and awarded summer funding supporting graduate research.

Undergraduate Research Assistant supported by Brian Dennison Memorial Award

UNIVERSITY OF NORTH CAROLINA AT ASHEVILLE

Asheville, NC USA

2018-2019

- Studied research methods, literature review, Python coding, and production of initial galaxy clustering measurements.

Publications

• 4 FIRST AUTHORED

Clontz, C., Seth, A.C., Wang, Z., [8 authors] (2026). *oMEGACat VIII. A Subpopulation Census of ω Centauri*. ApJ. 10.3847/1538-4357/ae3756.

Clontz, C., Seth, A.C., Wang, Z., [8 authors]. *oMEGACat V: Helium Enrichment in ω Centauri as a Function of Metallicity*. ApJ. 10.3847/1538-4357/ad1c5.

Clontz, C.; Seth, A.C.; Dotter, A.; [13 authors] (2024). *oMEGACat IV: Constraining Ages of Omega Centauri sub-giant branch stars with HST and MUSE*. ApJ. 10.3847/1538-4357/ad8621.

Clontz, C., Wake, D.A., Zheng Z. (2022). *The $H\alpha$ luminosity and stellar mass dependent clustering of star-forming galaxies at $0.7 < z < 1.5$ with 3D-HST*. MNRAS. 515. 2224C 10.1093/mnras/stac1889.

• 8 CO-AUTHORED

Souza, S.O., [3 authors], **Clontz, C.**, [13 authors] (2026). *oMEGACat. X. Shedding light on the disrupted dwarf galaxy of Omega Centauri*. arXiv. 10.48550/arXiv.2603.23589

Directed Chromosome diagram selection of P1 stars for analysis and comparison with stellar streams. Also provided scientific and editorial comments on the manuscript.

Wang, Z., Seth, A.C., **Clontz, C.**, [13 authors] (2026). *oMEGACat. IX. Chemical Tagging of Omega Centauri Populations with Machine-Learning-Inferred Abundances from the MUSE Spectrograph*. arXiv. 10.48550/arXiv.2603.01041

Directly involved in choosing how to best group and display results as they relate to the subpopulation generational indicators. Also provided scientific and editorial comments on the manuscript.

Wang, Z., [5 authors], **Clontz, C.**, [10 authors] (2025). *oMEGACat. VII. Tracing Interstellar and Intracluster Medium of ω Centauri using Sodium Absorptions*. ApJ. 10.3847/1538-4357/ae102c.

Provided scientific and editorial comments on manuscript.

Häberle, M., Neumayer, N., **Clontz, C.**, [15 authors] (2025). *oMEGACat VI: Analysis of the Overall Kinematics of Omega Centauri in 3D: Velocity Dispersion, Kinematic Distance, Anisotropy, and Energy Equipartition*. ApJ. 10.3847/1538-4357/adbe67.

Provided scientific and editorial comments on manuscript.

Häberle, M., [7 authors], **Clontz, C.**, [7 authors] (2024). *Fast-moving stars around an intermediate-mass black hole in Omega Centauri*. Nature volume 631, pages 285–288.

Contributed code for velocity dispersion calculation as well as provided scientific and editorial comments on manuscript.

Nitschai, M. S., Neumayer, N., Häberle, M., **Clontz, C.**, [14 authors] (2024). *oMEGACat III. Multi-band photometry and metallicities reveal spatially well-mixed populations within ω Centauri's half-light radius*. ApJ. 10.3847/1538-4357/ad5289.

Contributed to building of chromosome maps and provided comments on manuscript.

Häberle, M., Neumayer, N., [2 authors], **Clontz, C.**, [13 authors] (2024). *oMEGACat II – Photometry and proper motions for 1.4 million stars in Omega Centauri and its rotation in the plane of the sky*. ApJ. 10.3847/1538-4357/ad47f5.

Quality tested the photometric catalog, cross-matched with other catalogs, and contributed to development of differential reddening correction as well as provided scientific and editorial comments on manuscript.

Nitschai, M. S., Neumayer, N., **Clontz, C.**, [17 authors] (2023). *oMEGACat. I. MUSE Spectroscopy of 300,000 Stars within the Half-light Radius of ω Centauri*. *ApJ*. 10.3847/1538-4357/acf5db.

Contributed atomic diffusion correction and validation and provided scientific and editorial comments on manuscript.

Community Involvement and Service

July 2023 - Present	Sustainability Committee Coordinator , Max Planck Institute for Astronomy Organized invited speakers, improved bike infrastructure, established exchange cabinet, collected and disseminated community sustainability resources.	Heidelberg, Germany
May 2023 - Present	Steering Committee Member: Astronomers for Planet Earth , (A4E) Involved in monthly committee meetings, created non-profit chapter, new-member meetups, organizing virtual conferencing, climate education and resource dissemination.	Heidelberg, Germany
Oct. 2023 & April 2025	Astronomy Outreach , MPIA Open Day: Presented public-level astronomy talk + Girls' Day: 1-on-1 student mentoring	Heidelberg, Germany
2020-2022	PASSAGE Coordinator + Womens'+ Member , University of Utah + MPIA Organized monthly luncheons, film screenings, headshots, increased representation.	SLC & Heidelberg
2019-2021	Peer Mentor Program , University of Utah Provide support and resources to new graduate students through monthly meetings.	Salt Lake City, UT

Contributed Presentations at International Conferences

- 2026. Formation Scenarios for Complex Stellar Populations. Stellar Elemental Abundances Meeting, Uppsala University: Uppsala, Sweden
- 2026. Combining NSC Probes to Study Our Nearest Galaxy Center. Special Seminar, Lund University: Lund, Sweden
- 2025. Omega Centauri: Combining NSC assembly probes. Special Seminar, AIP: Potsdam, Germany
- 2025. A Subpopulation Census of Omega Centauri. Königstuhl Colloquium, MPIA: Heidelberg, Germany
- 2025. Omega Centauri: a unique probe of NSC assembly. Galactic Centers as Tracers of Galaxy Evolution: Leiden, NL
- 2025. Omega Cen: Combining NSC Assembly Probes [poster]. Bridging Scales: Matera, Italy
- 2025. Probing NSC Assembly: Omega Centauri. EAS 2025: Cork, Ireland
- 2025. The Formation History of Omega Cen. ESO Vitacura Headquarters: Santiago, Chile
- 2025. Omega Centauri: Our nearest Galactic Center. MPIA Galaxy Coffee: Heidelberg, Germany
- 2024. Science made possible by MUSE. A Decade of Discoveries with MUSE and Beyond: Garching, Germany
- 2024. Invited Talk: Probes of Omega Cen's Formation. University of Göttingen: Göttingen, Germany
- 2024. The Assembly History of Omega Centauri [poster]. The Milky Way Assembly Tale Conference: Bologna, Italy
- 2024. oMEGACat: Unraveling Omega Centauri. MPIA Galaxies and Cosmology Department Retreat: Schöntal, Germany
- 2023. The Formation History of the Nearest Galaxy Nucleus Omega Centauri. A multi-wavelength view on globular clusters near and far: from JWST to the ELT: Sexten, Italy
- 2023. The Formation of the Nearest Galaxy Nucleus Omega Centauri. GalPhases23: Phases of Galactic Evolution as Traced by Stellar Populations and Star Cluster: Sexten, Italy
- 2022. Omega Centauri Mono-metallic Subpopulation Ages. American Astronomical Society 240th Meeting: Pasadena, CA
- 2020. Halo Occupation Distribution (HOD) modeling of $H\alpha$ luminous galaxies at $0.7 < z < 1.5$ with 3D-HST: University of Utah Graduate Research Symposium: Salt Lake City, UT USA